



Comparative Examination of Master's Students' and Faculty Members' Maintenance of Academic Integrity in the Age of AI

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Abstract

The uncontrolled utilization of Artificial Intelligence (AI) tools among higher education students has incited debates about their ethical use, particularly in relation to academic integrity. This qualitative study closely investigates how students engage with AI tools in their academic work and the strategies they uphold to ensure and maintain ethical academic practices. Moreover, it explores how Moroccan university professors employ AI in their work and how they guarantee academic integrity while teaching and assessing students' performance. Reflective reports from fifty-one Moroccan master's students and twenty EFL professors were systematically coded and analyzed using thematic analysis. The findings of this study show that unethical and irresponsible use of AI tools can pose threats to academic integrity, including issues of plagiarism and academic dishonesty. Both students and professors acknowledge the wide risks associated with AI misuse and over-reliance on these tools, which can eventually undermine creativity and originality. Insights from this study, additionally, reveal the participants' advocacy for AI literacy programs and the need to rethink traditional assessment to assure ethical AI use in academia and fair evaluation of students' competencies. The findings of the present study contribute to the ongoing discussion on the risks and benefits of AI in education, accentuating the importance of developing institutional policies alongside AI governance and ethics programs in order to preserve academic integrity, while maximizing the pedagogical upsides of AI.

Keywords Artificial intelligence · Academic integrity · Ethical use · AI literacy

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Introduction and Review of the Literature

The progressive utilization of Large language models (LLMs) and Artificial Intelligence (AI) tools in education has proven to have major potential in changing how learners access information (Chowdhury & Chowdhury, 2024), thereby making it more accessible (Dziatkovskii, 2023). Additionally, personalized learning experiences are made possible by AI technologies, such as adaptive learning platforms and intelligent tutoring systems, which analyze students' data to customize learning to meet each student's needs (Borasi, 2024; Mishara, 2024). In spite of these benefits, students' engagements with these models are igniting critical debates related to their risks and ethical challenges, namely data security, algorithm bias, and the digital divide (Eden et al., 2024). Moreover, professors are becoming more and more apprehensive about students' use of AI tools and its impact on academic integrity, prompting concerns about the fairness of the assessment of students' performance (Eden et al., 2024). The generative capabilities of AI tools can be used to complete academic assignments and research projects among students, resulting in cheating, plagiarism, and academic misconduct (Dhruv et al., 2024; Khatri & Karki, 2023). Moreover, these tools can evade existing similarity-detection software, making it difficult to recognize AI-generated content (Akkaş et al., 2024; Denkin, 2024).

This compels professors to develop strategies to mitigate issues related to academic integrity in the age of AI. One widely adopted approach by both professors and institutions is using plagiarism detection software to verify the originality of students' work (Paul, 2024). Another recommended approach involves live assessments and competency-based evaluations, which require students to demonstrate their skills in real-time, reducing the opportunity for AI-assisted cheating (Fatima et al., 2024). Given its importance in maintaining the value of qualifications and endorsing ethical standards in educational settings (Balalle & Pannilage, 2025), academic integrity remains a key tenet that higher education professors need to proactively ensure and promote. Educators play a vital role in helping students to view AI as a supportive tool rather than a shortcut. Professors must emphasize the importance of critical analysis and the evaluation of credible sources when gathering information (Yeo, 2023). Moreover, by focusing on activities that encourage critical thinking and in-depth analysis, students are less likely to rely solely on AI tools, fostering their independent learning and problem-solving skills (Hosayny et al., 2025; Paraso et al., 2024).

AI literacy is increasingly recognized as a crucial competence that both professors and students must develop. Not only does AI literacy equip students with the skills needed to adapt to evolving technological demands of workforce (Ghani et al., 2024), but it also deepens their understanding of the ethical implications and limitations of AI, promoting a culture of honesty and ethical conduct in academic settings (Ganguly & Pandey, 2024; Khatri & Karki, 2023). The provision of early ongoing education on academic integrity is crucial (Tulin, 2024). As we continue to prioritize academic integrity within the AI landscape, institutions need to amend their definitions of academic misconduct to incorporate AI tools. By creating clear and accessible policies and guidelines, students will have a better chance of being aware of what is considered academic malpractice (Balalle & Pannilage, 2025; Paul, 2024).

It is also mandatory to create ethical frameworks regarding how AI will be utilized in higher education. These ethical frameworks need to reflect concepts of transparency, accountability, and responsible use of AI tools in the classroom (Gustilo et al., 2024; Per-

kins, 2023). In contrast, such policies and measures remain nascent and limited in the Moroccan context. Although discussions have surfaced about the implications of AI on academic practices (Ejjami, 2024; Moukhliiss et al., 2024), there are no yet concrete guidelines. Furthermore, there is a noticeable lack of qualitative studies exploring the perceptions of both students and faculty members regarding the issues to academic integrity in the age of AI, chiefly within Moroccan higher education institutions, particularly the strategies they employ to uphold it. In light of these concerns, the objectives of this study are threefold. This paper seeks to examine the ways Moroccan students and professors engage with AI in academic endeavors and how they view AI tools in relation to academic integrity. Moreover, it aims to explore the practices both students and university professors follow to preserve academic integrity in the age of AI, which is a shared responsibility between both parties. Lastly, this study elicits suggestions from both Moroccan professors and students with respect to the policies and measures that the university ought to adopt to address AI related academic integrity issues.

Therefore, this study aims to answer the three following research questions:

- How do Moroccan master's students and university professors view AI tools in relation to academic integrity?
- How do Moroccan master's students and university professors maintain academic integrity in the age of AI?
- How do master's students and university professors think universities should address AI related academic integrity issues?

Theoretical Framework

The present study adopted the Actor-Network Theory (ANT), which is originally developed by Bruno Latour, Michel Callon, and John Law in the late 1970s to early 1980s (O'Doherty, 2019). Originating from the field of science and technology studies, ANT has been integral in understanding how scientific facts are constructed through networks of human and non-human actors (Vibert, 2022). This theory views social systems as networks consisting of both non-human and human actors, which are dynamic and constantly evolving as actors form alliances and negotiate roles (Müller, 2015; Verhoeven, 2018).

ANT presents innovative methodological approaches that provide new perspectives to study social realities by focusing on the heterogeneity of agents and the relational dynamics within networks (Švantner, 2020). In the context of AI and education, ANT helps comprehend the influence that AI tools have on educational outcomes and practices, challenging traditional notions of teachers' and students' roles (Birkbak, 2023), where AI tools are not merely technical aids but active participants that shape educational practices (Fenwick & Edwards, 2010).

Applied to the issue of academic integrity, ANT in this study offers a broader perspective when exploring how AI tools interact with institutional rules, students' behaviors and faculty's perceptions to construct novel ways of learning and knowledge. Since ANT emphasizes the interplay between human actors such as students and professors, and non-human actors, like GenAI tools, it recommends that training on the ethical use of these tools should be a proactive part of educational strategies, calling for critical engagement with these technolo-

gies as co-constructors of academic work, transparency, and proper attribution (Crawford, 2020).

By analyzing how master's students alongside faculty members navigate academic responsibilities when using GenAI in academic endeavors, this theory illustrates how integrity is maintained, challenged, or redefined through a network of human to machine interactions. It thus offers a theoretical framework to this study to answer the research questions, enabling the assessment of the reconstitution of academic roles, responsibilities, and ethical considerations in the age of AI.

Methods

Research Design

This study adopts a qualitative research approach using a phenomenology design. The qualitative approach permits a comprehensive exploration of human experiences (Pitts & Price, 2022), guaranteeing the elicitation of deeper understandings into participants' perceptions and behaviors (Rahman, 2016). Phenomenology, in this study, serves to explore how master's and university professors experience the use of AI tools in their academic practices, while uncovering ethical challenges that these tools pose to academic honesty, which has become an integral concern in academic settings. Further, this design facilitates the acquisition of more profound insights into participants' perspectives on how universities should tackle AI related academic integrity. Particularly, this design is well-suited for exploring understudied phenomena such as ethical tensions and coping strategies related to the use of GenAI in academic work. Due to its focus on the everyday experiences of teachers and students, phenomenology offers a grounded approach that captures the lived realities of faculty members as well as students, leading to practical insights that are directly applicable to teaching practices and institutional policies in academic settings (Bonyadi, 2023).

Participants and Sampling

The target population of the current study comprises EFL Moroccan master's students and Moroccan university professors. The selection of master's students is justified by their extensive exposure and the engagement with academic writing process and their regular undertaking of academic assignments, namely writing reports and developing presentations. A convenience sample of fifty-one master's students was recruited at a public Moroccan university.

As for university professors, their participation in this study is of importance, since they play a key role in promoting and reserving academic integrity, particularly in the present moment where AI tools are widely used among students. In this study, twenty Moroccan EFL university professors, from the same faculty from which students were selected, participated and selected through convenience sampling. The sample included professors, ranging from full-time professors to a few part-time doctoral students who teach at the faculty. Each participant, both from students to professors, was provided with special codes or pseudonyms, to ensure strict anonymity. Further, to protect confidentiality, especially given

the small sample size that could make participants identifiable, both the name of the institution and the specific master's programs to which students belong to have been anonymised.

The difference in the number of participants between students versus professors is simply due to their varying availability and their readiness to participate in this research study. Given their professional commitments and time constraints, faculty members were difficult to coordinate with.

While convenience sampling is often criticized for its lack of generalizability, it is ethically advantageous as it respects voluntary participation and is less intrusive than some probability sampling (Suen et al., 2014), especially given the fact that AI integrity can be a sensitive and relatively subjective area to research. To mitigate any potential bias introduced by the convenience sampling, participants' diversity in terms of teaching experience, academic rank, and program affiliation was taken into consideration.

Data Collection Tool

To collect data from both students and professors, reflective reports were used. Generally, reflective practices are of value in educational practices, as they help foster a positive attitude towards learning and self-assessment and critical reflection, by recognizing strengths and weaknesses, hence upgrades the validity and richness of collected data (Heath, 2018). For students, reflective writing provided them with a private opportunity to describe their use of AI tools without the immediacy of a face-to-face interview and without social pressure, which informs the social desirability bias. Given the ethical challenges around academic integrity and what may be perceived as stigma associated with some of their AI related practices, some students would not want to speak frankly in an interview. For university professors, logistical considerations were central to the choice of this method. Synchronous interviews were hard to arrange with many professors, given their busy schedules. Reflective reports provided flexibility and increased efficiency for participants to engage with the research without having disrupting their professional commitments.

The reflective reports were semi-guided, consisting of five questions to both students and professors. The design was informed by a study conducted by Karkoulou et al. (2024), from which four reflective reports' questions were taken and adapted to better align with the current study's research questions. An additional question was added by the researcher to elicit participants' suggestions with regard to institutional policies and measures that universities are supposed to implement to address AI-related academic integrity issues.

Students' Questions

- 1 Do you use AI tools in your academic work (writing reports, doing assignments, preparing presentations, preparing for exams...)? if so, how do you use them?
- 2 Do AI tools pose a risk to maintain students' academic integrity? Why or why not?
- 3 What are the strategies that you use to ensure academic integrity and avoid plagiarism?
- 4 In your opinion, can AI tools serve as helpful tools rather than being a threat? Explain.
- 5 How do you think universities should address AI-related academic integrity issues? What policies or measures would you suggest?

Professors' Questions

- 1 Do you use AI tools in your work? If so, how do you use them?
- 2 Do AI tools pose a risk to maintain students' academic integrity? Why or why not?
- 3 How do you ensure academic integrity while teaching and assessing students' work in an era where AI tools can generate content for assignments?
- 4 In your opinion, can AI tools serve as helpful tools rather than being a threat? Explain.
- 5 How do you think universities should address AI- related academic integrity issues? What policies or measures would you suggest?

The two sets of reflective reports' questions tried to align with the roles of the respective participant groups, students and professors, while maintaining thematic consistency. Students' questions focus on their individual use of AI for academic tasks and how they demonstrate integrity whilst using AI tools. Conversely, professors' questions put emphasis on the use of AI for teaching and assessment and how they maintain academic integrity regarding students' work. Although both groups were asked more or less the same questions but their responses differ in a way that professors evaluate AI from an institutional and an evaluative perspective, whereas students reflect on their unique experiences with AI tools.

Procedures

Data collection was conducted over the course of approximately one week, from the 25th to the 30th of March. To reduce the likeliness of students using AI to complete the reflective reports, participants were handed physical copies of the reflective report prompts during their break and were asked to complete them by hand under supervision to discourage AI interference. On the other hand, reflective report prompts for university professors were sent via email to accommodate their busy schedules and ease accessibility. Instructors were explicitly requested to provide original reflections and share personal experiences and opinions as a way to minimize the likelihood of AI generated responses. Our institution granted ethical approval for this study and all participants provided informed consent prior to data collection. Furthermore, participants were informed that participation is entirely voluntary for both groups and no individual was coerced or obliged to take part in this study in any way. To ensure ethical conduct of research, participants were informed about the purpose of the study and their right to withdraw at any time.

To facilitate data management and analysis, the qualitative data collected from students was scanned and securely stored in a separate file from that of the professors' digital submissions. Following Braun and Clarke's (2006) six phases framework, the responses from both groups were subsequently thematically analyzed in order to identify emerging themes and recurring patterns. These phases include the following: (1) reading the data several times to familiarize oneself with it; (2) making preliminary codes; (3) searching for themes; (4) evaluating themes; (5) defining and labeling themes, and (6) producing the report. Furthermore, an inductive approach to coding was applied, allowing the themes to emerge from the data. Given that the researchers are affiliated with the institution where the study was conducted, they had access to participants and knowledge of the institutional culture. After independently coding the data, two coders convened to compare their analyses and discussed any

points of contention in order to reach a consensus on the final themes. This helped validate the credibility and added consistency to the findings. Finally, to avoid bias in the interpretation of data, journaling and peer debriefing were used to preserve analytical neutrality and guarantee that participants' voices remained at the forefront and central.

Results

An in-depth thematic analysis revealed five key themes arising from the data that showcase participants tendencies, perceptions and concerns related to the use of AI and academic integrity the predominate themes were:

- (1) AI Usage in Academic Work,
- (2) Benefits of AI
- (3) Threats of AI on Academic Integrity,
- (4) Strategies to Ensure Academic Integrity,
- (5) Institutional Policies and Measures.

The researcher will use a comparative lens to consider the responses from students and professors to actively demonstrate similarities and differences between their perspectives and practices.

AI Usage in Academic Work

The reflective reports indicated that master students mainly resort to AI tools to understand concepts, to summarize long assigned readings, which they described as overwhelming and stressful. In light of ANT, AI tools are surfacing as non-human actors that redirect academic activities by amending the liaisons between students, long texts and exams. For instance, IB stated that "I noticed when the overload becomes overwhelming, I use it [AI] more to summarize texts or to shorten my reading time". Furthermore, students tend to use AI tools to undertake exams revision by breaking complex concepts into bite size pieces of information, making them simpler and rememberable. RG explained that "I give the reading materials to AI software and write a prompt that will dissect it into parts that are readable and well detailed". This strategy enhances retention and makes the exam period much more manageable, thereby boosting students' confidence in dealing with large amounts of readings. That is, the non-human actors (AI tools) are strategically shaping the production of learning, making them active actant rather than passive instruments.

Regarding confidence, NB answered the first question about AI usage by this response reflects a rather negative perception of using AI tools in academic work. The term "unfortunately" implies a feeling of uneasiness and discomfort or an uncertainty regarding the implications of AI use.

Not only does AI help students during exam preparation, but it is also regarded as a more uncomplicated and more straightforward than using Google. LL described, "yes, I use AI in some of my assignments. I use AI to look for information that I can't find easily on Google". This statement suggests that AI tools are a user-friendly alternative compared to traditional research engines like Google. This exhibits a preference for AI tools among students.

Professors, on the other hand, actively incorporate AI in their work, particularly for lesson planning and generating interactive activities. As one professor explained:

Yes, I utilize AI tools for automating manual tasks. Particularly, I use them in lesson planning and drafting some thoughts. I also use them for fast searching for appropriate materials and activities to be incorporated in classes such as interactive activities when I am teaching English for STEM education or composition courses. (OEH)

Automating tasks using AI allows instructors to dedicate more time to the pedagogical aspects of their teaching. RA shared a similar standpoint, stating, “Yes, I use AI tools primarily to design and enrich my teaching materials. They help me create engaging language learning resources, generate ideas, and adapt texts to suit different proficiency levels”. RA further stresses the role of AI as an enabler and a complement rather than a substitute for the professors’ expertise, while maintaining key academic responsibilities by adding “I view AI as a supportive assistant, not a replacement, ensuring that my academic voice and pedagogical principles remain central”. Given this shared dependence on AI tools among students and professors, these non-human actors form an obligatory passage point in higher education contexts, where both learners and educators employ these technologies to meet their academic outcomes and objectives. Nevertheless, only two professors stated that they do not use AI tools in their work, highlighting a moment of network fragility. SK explained:

No. As far as I am concerned, AI is still a blurred concept. Apart from ChatGPT, I do not know any other platforms that we can use. In fact, whether any teachers use AI in their teaching practices is doubtful (SK).

Benefits of AI

The reflective reports’ findings additionally shed light on the perceived benefits of AI among students and professors. The majority of students consider AI use as not a mere source of information but as non-human actors, providing pedagogical guidance and are capable of advancing students’ academic achievements. In this network, AI tools further appear to be an obligatory passage point, since they offer personalized learning opportunities for students who are struggling. BK declared:

A lot of students struggle academically and are in need of personalized instruction. That is what I believe AI tools are designed and should be used for; to guide, assess and help students develop and sharpen their skills to advance academically (BK).

This statement raises broader implications of AI in promoting equity and inclusion in education to all kinds of students and bridging learning divides as well as promoting academic success. AE shared a similar opinion by stating that “I spent my first semester not using it [AI] and it put me behind my classmates, as reading assignments take so much time, while AI can help me understand in a fraction of the time”. AE’s statement portrays AI’s practical impact in allowing learners to keep pace with their peers by offering personalized and streamlined learning support.

Beyond its time saving benefits, some students reckon AI tools as offering further comprehensive and accessible learning experiences than professors, especially when it comes to explaining and clarifying complex concepts. ZB described:

Personally, they [AI tools] help me a lot to find connections between theories and concepts and author stances on certain issues. AI tools provide a fully comprehensive learning experience that we students may not get from professors, or just from searching the web. (ZB)

Sharing a similar opinion, FE stated that “I do believe that AI is an effective tool and has the potential to support students more than teachers themselves”. In accordance with ANT, the two statements show a critical reconfiguration of agency in this network, where human and non-human actors have equal analytical status, uniform with ‘flat ontology’. However, this challenged dichotomy of agency and structure exposes a moment of fragility, wherein authority is ceased to be monopolized by educators but it is negotiated between human actors (professors) and non-human actors (AI tools).

By the same token, professors, like students, regard AI-powered technologies as an advantageous source that supports both learning, teaching and giving immediate feedback to students, which has been proven if used ethically and responsibly among students. IS explained, “AI can support students in improving their writing, organizing their ideas, and accessing information quickly”. RA further added, “AI can be a valuable educational tool when used ethically. It supports personalized learning, aids diverse learners, and enhances feedback”.

Threats of AI on Academic Integrity

The findings revealed that students exhibit an explicit recognition of the downsides of AI use, most notably regarding issues of academic integrity. A recurring concern voiced by students is that overreliance on AI tools can stifle the sense of originality and creativity in their academic endeavors, thereby jeopardizing the authenticity and originality of their work. For example, YB stated:

I do see AI tools as a threat to academic integrity. The reason for this, is that students are not willing to put their personal effort into making or writing original work of their own. (YB)

They, additionally, expressed apprehension that frequent use of AI tools has the ability to hinder cognitive efforts, namely the vital critical thinking skills, which may affect students' diligence in the long term, by becoming habituated to shortcuts rather than deeply engaging with their academic learning. BZ explained:

Yes I do believe that over reliance on AI tools can stand as a hindrance to the development of students' critical thinking, creativity, and diligence, they will get used to shortcuts to accomplish their work (BZ).

Students further encapsulated the main concerns of academic dishonesty that are cheating and plagiarism. For instance, RL stated that “AI can be viewed as a threat if it’s used for cheating and plagiarism”. Building on this concern, ZB highlighted that as tools become more and more sophisticated, it created a challenge for educators to accurately assess students’ authentic efforts and competence. ZB explained, “With the ongoing development these tools may pose a challenge for educators to evaluate students’ actual efforts and competence”. Moreover, ZB added, “Using AI tools for example in research creates less credible students”. This statement implies that AI-powered technologies usage has long-term implications for students’ academic growth, suggesting that the use of AI without intentional and critical thought may ultimately subside research skills as well as academic credibility.

Among professors, perspectives differ. Some participants attributed drawbacks to students’ usage choices rather than to the tools themselves. That is to say, AI tools pose a threat to academic integrity solely when used passively and without critical engagement. RA explained:

For instance, autonomous and motivated learners are more likely to use AI tools to enhance their understanding, refine their ideas, or seek language support without compromising their originality or critical thinking. In contrast, passive or disengaged students may misuse AI by copying entire assignments without personal input or reflection. Therefore, the real threat lies not in the tool itself, but in the way it is used and the values instilled in students regarding ethical academic practices (RA).

Sharing similar opinion, CO asserted that “AI is a threat if there is not sufficient knowledge on AI ethics”, affirming that risk associated with AI tools are not necessarily related to the technologies themselves, but rather from the lack of adequate training and awareness of how they should be used: responsibly and ethically.

In response to the fourth reflective question, whether AI tools have the potential to become a helpful tool instead of a threat, HE firmly expressed their pessimistic stance, stating, “Absolutely not. While AI can be helpful, its potential risks WILL soon outweigh the benefits”. Although HE acknowledges the utility of AI but emphasizes the negative consequences that are more likely to surpass its advantages. In effect, the fact that AI’s agency and byproducts are neither predetermined nor fixed but rather arise via the participatory connections with human actors, institutional norms, and ethical rules showcases the contingent and unstable nature of the actor-network. Academic integrity and critical engagement are, thus, not permanent characteristics of students or AI tools themselves, but rather incipient results of the connections among the actants of the network.

Strategies To Ensure Academic Integrity

Students’ reflective reports confirmed professors’ observations that students are often not knowledgeable enough about AI ethics, revealing ongoing translation processes but they remain incomplete in the network of academic integrity. The reflective reports’ findings revealed a concerning trend among a big number of students, admitting that they take very few, if any, academic integrity measures in using AI tools. Students do not fully understand the ethical implications of AI use and do not see the relevance of engaging critically with AI generated content. For example, FD stated, “Unfortunately, I do not take any measures

yet, since I do not have enough knowledge on the topic”. LB added, “Often, I do not check whether my work is plagiarized. Maybe that’s something I should consider from now on”. AE also stated, “ I don’t make any efforts in that regard”. This signals a weak enrollment of learners, as they are not entirely aligned with the network of ethical AI use.

However, some students demonstrated some conscious efforts and strategies to maintain academic integrity when using AI. For instance, several students reported adopting strategies such as using their proper words and avoiding copying and pasting the AI generated content. MK articulated:

I ensure that by making sure to write on my own proper style and not copy and paste the answers. I also try my best to be critical when gathering information from AI and of course conserve my own reasoning. (MK)

Paraphrasing is another widely mentioned tactic used among students to avoid plagiarism. For instance, AB stated that “ I do use my own paraphrasing to change words and expressions”. ZT added “frankly speaking, I only paraphrase the generated texts”. However, ZL sheds light upon a strategic yet ethically ambiguous practice to bypass AI detection tools, which is using other AI tools. While this approach may circumvent some of the technical issues related to plagiarism, it certainly raises several ethical issues. ZL mentioned:

Honestly, I use other AI tools to paraphrase AI generated content, so when detect them using tools like GPTZero, you find it a 100% human generated content (ZL).

An additional method adopted by students is checking the credibility of AI generated content, resorting to books and articles, which is a more cautious approach. For instance, BK explained, “I check if the information provided is correct from articles and books”. IB added “ I do not rely on the generated information due to them either being plagiarized without sources or them simply being false”. IB’s statement shows a critical awareness of the ingrained limitation of AI, particularly its tendency to generate false content or false citations.

One final strategy shared by students to ensure academic integrity involves using Generative AI (GenAI) tools to search for resources, rather than solely relying on the generated content. CA highlighted the “search on web” feature, which allows users to access resources directly from the internet. CA stated, “there is an option of “search on web” in ChatGPT that helps in not falling into plagiarism by providing resources”. Similarly, ZB underlined their proactive approach by stating, “I always prompt it [AI] to provide citations and corresponding resources”.

In their efforts to ensure academic honesty, professors draw on multiple, ranging from sensitizing students to engage ethically with AI to redesigning learning tasks to curb AI misuse. Attempting to stabilize the network, OEH suggested fostering the habit of properly citing the resources and to reinforce the importance of critical scrutiny of AI generated content. To illustrate OHE stated:

For instance when giving students writing tasks, I inform students to cite works they have consulted or used in their essays even if its AI generated it should be mentioned. This procedure would initially teach them to maintain academic integrity in citing

what is not theirs and reporting only their own ideas and works. Then later on, I start instructing students on how to use the tool as an assistant and not take anything from it for granted in order to maintain originality and academic integrity (OEH).

AS advocated for a focus on problems- solving tasks in order to engage students deeply in their learning. AS described, “one would need to completely change what they assign to their students. Perhaps more focus on problem-solving tasks would do the trick”. Expressing a parallel viewpoint, AB encourages critical use of AI by encouraging debates, comparing human and AI generated assignments and assessing the strengths and weaknesses of AI. Quoting the participant:

I focus more on students, whose awareness and maturity are decisive in the last instance of the kind of AI use that will take place. As such, I occasionally invoke debates on this issue and invite students to discuss. There were times when I compared the human version and AI version of a written assignment and emphasized AI's pitfalls and biases (AB).

Addressing AI use in academic setting remain a complex endeavor, particularly identifying AI generated content. Some professors propose using AI detection software to discourage AI misuse among students, as SK suggested:

It is really hard; however, I think we, as teachers, should warn students against the negative effects that AI can have on their total academic performance, and hence on their future professional life. We should also make them aware that there are software that can tell us whether the student used AI or not (SK).

In opposition, other professors expressed skepticism about the potency of these AI detection tools, which are reported to be unreliable and inaccurate. For example, AB declared that “I don’t think that AI-detection tools currently available are the solution for they are still inaccurate and thus unreliable”. Generally, the interrelated human and non-human actors make the network of academic integrity unstable and partially translated in this case. That is to say, academic integrity is not a fixed attribute, but it is produced through continuous negotiation and incomplete translations between the diverse actants within the network.

Institutional Policies and Measures

In terms of translation within the academic integrity network, there is a shared consensus amongst students and professors with respect to the measures and policies that universities should implement to address AI related academic integrity matters. The majority of students and professors stressed the promotion of AI literacy instead of the complete ban of AI in education, exhibiting early stages of interestment. For example, a student recommended the introduction of modules to equip students not only with technical skills but also with a deep grasp of ethical implications. Quoting the student:

I think universities should include some modules about the use of AI so that students can use it effectively, and they will know about different ethical issues. Universities should promote an environment of creativity and critical thinking (AR).

Echoing similar perspective, LB urged the implementation of AI tools in the educational system, while offering courses on the proper use of AI for students. LB stated that, “universities must focus on incorporating AI tools into the education system. They must also consider teaching ethical ways through which students could use AI without any harm”.

However, several students advocated for a rather strict approach, which is punishment and the ban of AI use particularly during exams. For instance, BK mentioned, “if any of the introduced instruction were to be unfollowed by the students then some form of measures or punishment should be taken”. Moreover, BZ stated “during exams AI use should be banned”.

Other students underscored the necessity of using sophisticated AI and plagiarism detection systems to uphold academic integrity in the era of AI. As an example, AK stated, “I suggest opting for an advanced system that detects plagiarism for example”.

Collectively, if not all, the majority of faculty members evoked practical measures that educational institutions need to enforce to ration the unrestricted use of AI tools in education, particularly among students. First of these measures is the introduction of digital literacy initiatives, namely through undertaking workshops and offering dedicated modules to students and professors together. For example, HL explained, “universities should offer workshops and integrate digital literacy into curricula to help students understand how to use AI ethically and responsibly”. Professors further alluded to the reconsideration of assessment and the promotion of open dialogue to foster the culture of academic honesty in educational settings. For example, AE stated, “assessment in the era of AI should focus on critical thinking, which is an aspect that is purely human and that can be enhanced by AI-assisted tools”. HL further added:

Assessments should be redesigned to focus more on critical thinking, creativity, and in-class or oral tasks that are harder to outsource to AI. Finally, promoting a culture of academic honesty and open dialogue about the role of AI in education will help students make informed and ethical choices. (HL)

In addition to promoting AI literacy and ethics, professors join students and they suggested that there should be clear regulation and policies to attenuate the risk of AI misuse. For instance, MA stated, “they [universities] should write many regulations outlining how students should use them, how they aid them, and how they add a unique touch”. Moreover, RA recommended, “Develop clear, updated academic integrity policies that specifically address AI use—clarifying what constitutes acceptable and unacceptable practices”.

One professor, however, brought up a critical point regarding the feasibility of institutional reforms, highlighting the persistence of incomplete translation, expressing doubts about the system's capability to adapt to this swift technological advancement.

AS explained:

Organizations and establishments that operate on a massive scale are notorious for being slow to adapt to change. Moroccan universities are no exception. They exist in a

system rife with corruption, bureaucratic obstacles, and basic problems, such as overcrowding. How can one expect them to adapt to the age of AI, when many universities can't even properly use projectors? (AS)

In light of these limitations, AS advocated for a more individual approach, inviting other professors to take ownership to adapt to these technological changes. Suggesting educator-led change, AS asserted: “The best method for adapting is to start from one’s own classroom without waiting for the system to catch up because it likely won’t”.

Table 1 synthesizes the main themes extracted from the data.

Discussion

Generally, the findings of this study provided insights into the changing role of AI in Moroccan academic settings. Both EFL Moroccan master’s students and professors demonstrate an understanding of AI’s potential, acknowledging its upsides while also expressing concerns over the implications for academic integrity.

The majority of students reported using AI tools mainly to summarize overwhelming readings and get personalized learning and support to keep up with their peers. This is supported by Jumriah et al. (2024) findings, who revealed that the integration of AI in education facilitates better comprehension of engagement with lengthy tests, making it easier for students to summarize key points effectively. Moreover, students reported AI tools to be faster and contextually appropriate, making it particularly evident that they believed that AI might replace conventional engines like Google. However, this perspective is not entirely accurate, reflecting a surface-level understanding of how AI retrieves and processes data. As noted by Chowdhury et al. (2024), despite the rise of AI-powered tools, Google remains essential for information retrieval.

A notable finding was students’ perception of AI tools to be valuable contributors to equity and inclusion in education. For instance, a student stated that “I spent my first semester not using it [AI] and it put me behind my classmates, as reading assignments take so much time, while AI can help me understand in a fraction of the time”. This view aligns with those of Pawar and Khose (2024), who revealed that AI has the potential to promote equity by addressing education disparities and creating inclusive environments for diverse learners. However, without critical and structured engagement, learners’ reliance on AI can gradually foster overreliance, where these tools start to replace rather than enhance learning processes. As stated by BZ, a student:

Yes I do believe that over reliance on AI tools can stand as a hindrance to the development of students’ critical thinking, creativity, and diligence, they will get used to shortcuts to accomplish their work (BZ).

A similar stance was shared by Ayala-Pazmiño (2023) and Hosayny et al. (2025), revealing that students who overly rely on AI risk the reduction of their critical thinking and analytical skills, as such tools may encourage rote learning over deeper understanding.

Table 1 Summary of the five themes

Theme	Students' Perspectives	Professors' Perspectives
AI Usage in Academic Work	- Use AI to summarize readings and simplify complex content for exam revisions - View AI as more effective than traditional tools like Google. - Some discomfort with using AI.	- Use AI for lesson planning, generating materials and automation. - Emphasize AI as a “supportive assistant”, not as a replacement - Some professors lack awareness or experience with AI
Benefits of AI	- Enhances understanding and academic pace. - More accessible than professors or Google. - Enables personalized learning and inclusivity.	- Aids personalized and fast feedback for students. - Supports learners and enhances pedagogical practice when used ethically.
Threats of AI on Academic Integrity	- Fear of overreliance, loss of originality and reduced critical thinking. - Concerns over AI assisted cheating and credibility loss.	- Emphasize misuse due to lack of ethics training, not the tool itself. - Some professors are skeptical of AI's long term benefits.
Strategies to Ensure Academic Integrity	- Many students lack awareness of ethical use. - Some students paraphrase AI generated content without citing. - Some students critically assess AI generated content. - Other students use tools to bypass detection	- Some professors call for responsible use of AI - Promote citation practices, ethical engagement and task redesign (debates, comparisons) - Split on AI detection tools (some support it, others doubt its accuracy)
Institutional Policies and Measures	- Call for AI literacy modules and ethical training. - Some support bans in exams and stronger similarity detection.	- Support workshops, curriculum integration and clear AI policies - Advocate redesigning assessment and focus on ethical reasoning - Skepticism about system readiness

From the instructor's perspective, AI tools hold the capacity to streamline pedagogical planning, such as lesson planning and activity generation. This has been stated by a professor:

I mainly use it to help organize course materials. I would use it to generate practice tasks based on the covered materials or generate examples for rules, etc.(CO).

These findings support the growing literature on the utilization of AI by students and professors alike, especially using AI tools for generating individualized content, making lesson plans (Pender et al., 2022). However, while such practices increase productivity, this statement raises questions about the potential risks of AI influencing and forming pedagogical decisions. Indicated by Ayala-Pazmiño (2023), teachers' reliance on AI for decision-making may compromise their role of critically guiding and mentoring students. Reflecting this concern, a professor stated, "I view AI as a supportive assistant, not a replacement, ensuring that my academic voice and pedagogical principles remain central."

Beyond pedagogical planning, participants conveyed further drawbacks of AI, notably in relation to academic integrity. Students shared concerns that the use of AI tools could lead to plagiarism and cheating. For instance, BB mentioned, "overlying on AI is a problem that should be addressed, especially regarding academic integrity, because some students use it to cheat and copy answers and end up having better marks than other students who actually worked hard". A professor added:

I think many students have become lazy because of technology in general. They used to rely heavily on the Internet and now on ChatGPT. When correcting students' composition exams, I once found two identical essays: 'A trip to Marrakech', which could not have been produced by S3 students (SK).

These observations resonate with Sozon et al. (2024), revealing that AI tools, namely ChatGPT, have been misused by students, resulting in an increased instances of academic dishonesty, including cheating and plagiarism. Nonetheless, the claims about students becoming 'lazy because of technology' may not be the primary reason for their lack of motivation or effort. Instead, it may be the result of unclear assessment expectations and inadequate digital ethics training that could lead to AI reliance, as mentioned by Selwyn (2024).

This perspective is further reinforced by students' own reflections, which emphasize that excessive AI use can dull their creativity and weaken their cognitive skills. For example, CA stated, "it [AI] damages our abilities of doing things". These perspectives are also reinforced by several studies indicating a notable decline in cognitive effort over the past decade, with individuals exerting 20% less mental effort on daily tasks: a trend attributed to the increased reliance on technology (George et al., 2024).

Professors held comparable views to those of students, adding a crucial insight which is AI misuse often stems from the lack of digital literacy and ethical awareness. This finding makes parallel to the one of Mackay (2022) and Chávez et al. (2023), who concluded that students who enter higher education institutions without a proper understanding of digital literacy and the ethical use of AI in academic work tend to misemploy AI technologies. As one student expressed, "Unfortunately, the answer is yes I do [use AI]". The hesitant and uneasy tone in which the student admitted to using AI conveys a sense of apprehension and underlying shame, which is likely stemming from their inadequate understanding of AI ethics.

The answers to the question about strategies students apply to uphold academic integrity were largely indifferent and relatively relaxed regarding the ethical implications of using AI, with particular regard to plagiarism. This can be credited to the gap between students' and faculty members' comprehension of academic integrity (Murraya & Myersb, 2024). However, a few students stated that they verify the credibility of the AI-generated content by

crosschecking it with books and scholarly articles to preserve academic integrity, recognizing that AI tools can intermittently hallucinate and produce inaccurate information, misleading students. To illustrate, IB expressed “I do not rely on the generated information due to them either being plagiarized without sources or them simply being false”. This is aligned with Kamel (2024) findings, in which he emphasized that hallucinations and inaccuracies can lead to misinformation and biases.

While a few students reported critical engagement with AI-generated content, this behavior was not the norm. Another strategy that students use to maintain academic integrity is by paraphrasing AI-generated content without any mention or citation of the AI tools, which is still regarded as plagiarism. For instance, this student mentioned, “frankly speaking, I only paraphrase the generated texts”. This practice reveals a knowledge gap among students, as many seem to be oblivious that paraphrasing without giving credit remains against academic integrity standards, even when the source is an AI tool. Sysoyev (2024) supports this finding, attributing the way AI-generated content is seen as less unethical to plagiarize among students to the novelty of AI technologies and apparent ownership discrepancies they create. Such practices are often driven by the lack of systematic understanding of how to legally use GenAI among students, as noted by Simpson (2023). This further aligns with ANT's assertion that networks become fragile or unstable when the alignment between actors is weak or underdeveloped (Andrade & Urquhart, 2010). In this case, the lack of policy guidance, a non-human actor, adds to the ambiguity and confusion in students' perceptions of academic integrity.

Instructors expressed the striking need for open discussion to promote critical use of AI amongst students. As professor FA expressed, “We should raise the awareness of students with regard to the ethical use of AI”. This perspective is supported by several studies that recognized the prominence of open discussion on AI ethics. For instance, a study by Owino and Paschal (2023) indicates that special courses on the ethics of AI have the potential to aid students alongside researchers understand AI technologies' practical and ethical aspects, helping in fostering critical thinking and information evaluation skills. Though commendable in intent, this perspective warrants deeper scrutiny regarding its practical implementation, particularly about the extent to which educators themselves are prepared to lead such conversations. For instance, professor SK stated, “As far as I am concerned, AI is still a blurred concept”, pointing out the discrepancy between the call for ethical AI debates and professors' own confidence and competence using these technologies. Kizilcec (2023) and Yankova (2024) underscore the importance of taking into account social, psychological, and cultural aspects, which influence how educators see, trust and adopt AI, to guarantee their openness and readiness to implement AI technologies into education.

The issue of AI detectors was met with mixed views among professors. Some believed that they could help in ensuring academic authenticity, whereas others questioned their reliability. Sadasivan et al. (2023) support our findings, disclosing that AI detectors are vulnerable to simple paraphrasing, in which text produced by AI is slightly altered, can bypass detection, making it challenging.

Finally, the last question of the reflective report explored the policies and measures institutions could carry out to combat academic integrity issues. Together with students, professors called for the need to introduce AI literacy and ethics courses with a view to training students on the appropriate ways of AI use, maximizing its benefits while minimizing its risks. For instance, a professor stated, “Advocating AI literacy, which is very important”.

Participants also recommended the undertaking of workshops and seminars as efficient avenues for sensitizing and promoting responsible academic practices. Consistent outcomes by (Rasky, 2023) yielded analogous results, revealing that workshops and reflective discussions can increase learners' understanding of AI, compelling them to use these tools ethically. The ANT explains how academic integrity is not static but rather co-constructed through interactions among students, faculty and AI tools. Moreover, ANT implicates a network that inherently works better when actors go through a process of translation to identify and co-align their goals and actions. (Tietjen, 2018). In this case, translation might include a combination of co-creating clear policies, co-training and institutional support in order to develop a shared understanding of the role of AI in academic work, as it is supported by several studies (Atenas et al., 2020; Königs et al., 2021; Mahmud, 2024).

Authentic and continuous assessment strategies, namely debates that favor creativity and in class tasks, were a further recommendation that professors put forward to endorse academic integrity in the age of AI. As stated by a professor:

I design assessments that emphasize critical analysis, personal reflection, and classroom engagement, making it harder to rely solely on AI-generated content. Third, I incorporate oral presentations and follow-up questions to verify students' understanding of their own work. (HL)

Parallel conclusions were drawn in a study conducted by Pramjeeth and Ramgovind (2024), which shows that authentic assessments that are focused on higher-order cognitive skills and problem-solving are less likely to be compromise by AI-supported malpractice. These assessments align with how pedagogy is devolving in the 21 st century.

In general, educators appear willing to engage in critical discourse regarding AI technologies; however, their successful participation requires ongoing institutional commitment and contextualized professional development to guarantee that they are satisfactorily supplied with the requisite knowledge and pedagogical competencies. This is in line with research by Mah and Groß (2024), who revealed that professors must have systematic professional development to address AI literacy; therefore, be able to use these technologies with confidence and responsibility.

Academic integrity in higher education is undergoing transformation, as emerging technologies, which are not merely passive aids but co-constructors, are actively reshaping academic practices as well as responsibilities. According to ANT, this is a reconfiguration of an academic network, where the traditional roles of professor and students are being negotiated. As supported by Kumar and Tissenbaum (2022), these technologies proactively redirect agency within learning spaces and shift the nature of hierarchy in a way that compels institutions to reevaluate their definitions of academic integrity, responsibility and authority are collectively constructed in the age of AI.

Conclusion and Implications

The purpose of this study was to explore the way EFL Moroccan masters' students and EFL professors navigate academic integrity in the era of AI as well as elicit their suggestions for institutional policies and guidelines to approach academic integrity issues. The findings of

this study provided insights into both students' and professors' engagement with AI tools in their academic practices. Participants reported that AI tools are invaluable in supporting individualized learning by assisting students during exam preparation through summarizing extensive readings, thereby making the learning process more efficient and effective. Moreover, professors primarily reported utilizing AI in designing activities and lesson planning.

Furthermore, the data showed that students along with professors, shared awareness of the potential drawbacks of AI on students' creativity, critical thinking skills and authenticity. Concerns about cheating and plagiarism were also expressed, which are reportedly ascribed to the insufficient training on AI ethics and responsible use of AI. This showed a rather lenient attitude towards maintaining academic integrity among students. Professors, on the other hand, reported to stimulate critical engagement of students with AI through debates and project based activities. The data also revealed that students and professors underlined the need for AI literacy initiatives to sensitize students on how to use these tools efficiently. Additionally, there was an urgent call for reconsidering traditional assessment in favor of more authentic approaches.

In light of these insights, this study puts forward recommendations for instructors, curriculum designers, and researchers. To ensure fair and accurate evaluation of students' competencies, instructors are urged to take personal initiatives by adopting more authentic assessment strategies. Educators can foster authentic learning, assessment practices through placing greater emphasis on continuous assessment and process-based assignments as they reflect students' creativity and uniqueness better than tradition assessment. Examples include designing portfolios, organizing oral defenses, supervising research-based projects and simulating conference panel presentations. Furthermore, faculty members ought to nurture open dialogue with students through, for instance, formal debates or informal class discussions, reflecting on the boundaries of AI and speculate on what academic integrity means in practice.

For curriculum designers, there is a clear need to incorporate AI literacy and digital ethics early on into higher education to assist students in acquiring and developing necessary skills to be conscientious users of AI tools. These programs can include activities that foster critical awareness of the weaknesses of GenAI output. For example, this could involve training students to reevaluate AI generated content for accuracy and reliability and to locate more reliable scholarly sources. In particular, students need to understand what constitutes plagiarism and its types and how to avoid it. Regarding researchers, while exciting work has explored rethinking assessment practices and introducing AI literacy programs, future research can move toward longitudinal studies that examine the long-term impact of these programs and conduct comparative trials evaluating the efficacy of AI detection tools versus redesigning assessment strategies.

This study is not without limitations. Firstly, this study limited itself to a single institution and did not include other faculties or universities of Morocco, which may have different institutional cultures regarding AI use and varied levels of digital literacy among faculty members and students. Secondly, this study depended on a small convenience sample of master's students, which may not capture the diversity of experiences and viewpoints that are found in a larger sample of students and professors; therefore, this sample strategy limits the generalizability of the findings. Thirdly, as a key data collection instrument, reflective reports pose potential for self-report bias. Students may want to represent themselves in a more favorable manner, particularly when reporting about an ethically charged topic such

as the use of AI and academic integrity. Lastly, in spite of the measures taken to ensure authenticity, there was still potential for some of the participants to have relied on AI tools when composing their reflective reports.

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Declarations

Ethical Approval and Consent To Participate The present study received approval from our institution of Moulay Ismail University and informed consent to participate was obtained from all participants before the commencement of the study.

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